

Administracao e Otimizacao de Bancos de Dados

Vetoriais para Aplicacoes de Inteligencia

Artificial usando PostgreSQL (pgvector)

Equipe: Jose Inamar de Medeiros Junior

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Fundamentação Teórica dos Bancos Vetoriais

No âmbito do aprendizado de máquina contemporâneo, a representação de dados não estruturados fundamenta-se no mapeamento de conceitos semânticos em espaços vetoriais contínuos de alta dimensionalidade¹. Matematicamente, um **embedding** é uma função de projeção $f: \mathcal{X} \rightarrow \mathbb{R}^d$, onde um objeto arbitrário $x \in \mathcal{X}$ (como um texto, uma imagem ou um áudio) é codificado em um vetor de números reais com dimensionalidade d tipicamente compreendida entre 10^2 e 10^4 ¹. Geometricamente, o vetor resultante $f(x)$ representa coordenadas em um espaço de Hilbert de alta dimensão, onde a proximidade geométrica entre dois pontos reflete a similaridade semântica dos objetos originais³.

Distância L2 (Euclidiana)

A Distância L2, ou Euclidiana, calcula o comprimento do segmento de reta que une dois pontos no espaço vetorial³. Matematicamente, é expressa pela raiz quadrada da soma das diferenças quadradas entre as coordenadas correspondentes de dois vetores $\mathbf{x}$ e $\mathbf{y}$:

Produto Escalar (Inner Product)

O Produto Escalar mede a projeção de um vetor sobre o outro, ponderada pelas magnitudes de ambos³. A sua expressão matemática é dada pelo somatório do produto das componentes correspondentes:

Similaridade de Cosseno

A Similaridade de Cosseno avalia exclusivamente o alinhamento angular entre dois vetores, neutralizando quaisquer disparidades introduzidas por suas magnitudes⁷. Geometricamente, calcula o cosseno do ângulo  formado entre as direções dos vetores no espaço multidimensional⁷. O pgvector calcula a Distância de Cosseno, definida como o complemento da similaridade de cosseno em relação à unidade⁸:

Métrica de Distância | Operador SQL | Intervalo de Saída | Sensibilidade à Magnitude | Caso de Uso Primário em IA

A Maldição da Dimensionalidade e a Intrinsecabilidade do Espaço Vetorial

Ao administrar sistemas de busca vetorial, o arquiteto de dados confronta-se com a maldição da dimensionalidade¹. Em espaços métricos de alta dimensão, a distância entre o ponto mais próximo e o mais distante converge para uma diferença marginal, tornando o particionamento espacial tradicional ineficiente. Pesquisas empíricas revelam que a revocação de algoritmos aproximados como o HNSW está diretamente atrelada à Dimensão Intrínseca Local (LID \- *Local Intrinsic Dimensionality*) dos vetores¹⁰.

Arquitetura do SGBD: PostgreSQL \+ pgvector

Justificativa Técnica da Escolha Arquitetural

A consolidação de infraestruturas de Inteligência Artificial motivou o surgimento de bancos de dados vetoriais dedicados e proprietários, como Pinecone, Milvus ou Qdrant¹¹. Contudo, a adoção do PostgreSQL estendido com o pgvector fundamenta-se em princípios rigorosos de engenharia de software e teoria de bancos de dados, oferecendo vantagens estratégicas incontornáveis¹.

Critério de Comparação | PostgreSQL \+ pgvector | SGBDs Vetoriais Dedicados

A fragmentação arquitetônica inerente ao uso de um banco de dados relacional clássico em conjunto com um motor vetorial especializado introduz riscos de sincronização¹¹. Quando uma entidade de negócio é atualizada, a modificação correspondente no banco vetorial ocorre de forma assíncrona, criando janelas de inconsistência em que dados desatualizados são expostos a pipelines de RAG¹¹. No PostgreSQL, as atualizações relacionais e a persistência de novos vetores de embedding ocorrem sob a égide da mesma transação ACID, eliminando falhas de gravação parcial¹³.

Gerenciamento de Armazenamento Físico de Dados Vetoriais

O PostgreSQL opera nativamente com um modelo de armazenamento físico baseado em páginas de tamanho fixo de 8 KB¹⁴. Toda tupla gravada em uma tabela deve ser estruturada de forma a respeitar esse limite ou delegar o excedente para mecanismos de transbordo¹⁴.

C

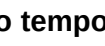
A partir desse leiaute físico, o espaço em bytes consumido exclusivamente pelos dados do vetor em disco é calculado como:

*** O tipo padrão vector (baseado em floats de 32 bits) está limitado a no máximo 2.000 dimensões para fins de indexação.**

O Conceito de Busca Híbrida (Hybrid Search)

A essência operacional do pgvector consiste na capacidade de unificar pesquisas estruturadas relacionais e buscas aproximadas por similaridade vetorial em uma única expressão SQL compilável⁹. No entanto, o otimizador de consultas do PostgreSQL precisa balancear o custo de execução de filtros de atributos contra o custo de travessia do grafo vetorial²⁸.

Planejamento de Filtragem via Índices Relacionais (Pre-filtering)

Se o filtro relacional (por exemplo, `WHERE tenant_id != '...'`) apresentar alta seletividade, retornando um número reduzido de linhas, o planejador opta por realizar um escaneamento de índice tradicional (B-Tree ou Hash)²⁸. As tuplas resultantes são filtradas em memória e o cálculo de distância vetorial é computado de forma exata (K-NN linear) apenas sobre este subconjunto qualificado, ignorando completamente o índice vetorial²¹. Esta abordagem garante  de revocação e baixo tempo de resposta sob alta seletividade²¹.

Planejamento de Escaneamento Iterativo do Índice (Iterative Index Scan)

Quando os filtros relacionais possuem baixa seletividade, ou seja, qualificam uma parcela substancial da tabela, o custo de avaliar linearmente os vetores torna-se proibitivo²⁸. O PostgreSQL decide utilizar o índice vetorial (HNSW ou IVFFlat) para guiar a recuperação⁹.

*** off: Desativa o comportamento iterativo, reproduzindo a pós-filtragem estática das versões anteriores (maior velocidade, alto risco de superfiltragem)³¹.**

Algoritmos de Indexação e Administração Física

Funcionamento Interno do IVFFlat e HNSW

Os algoritmos de indexação de aproximação (ANN) implementados no pgvector oferecem soluções distintas para o compromisso clássico entre velocidade de busca, tempo de compilação e custo de armazenamento³⁵.

Camada 2 (Esparsa \- Saltos Largos)

A velocidade do HNSW reside nesta estrutura multinível³¹. A busca inicia-se na camada mais alta por meio de uma busca gananciosa (**greedy search**), onde o algoritmo calcula a distância do vetor de consulta em relação aos vizinhos conectados ao ponto de entrada ativo⁴¹.

Conectividade do HNSW e Parâmetros Críticos de Construção

A qualidade topológica do grafo HNSW é governada diretamente pelos parâmetros configurados no momento de sua criação⁵:

* **m** (Número máximo de conexões bidirecionais por elemento): Define o limite máximo de arestas que cada nó pode estabelecer com vizinhos em cada camada do índice⁵. A camada base (Camada 0) constitui uma exceção estrutural fundamental: ela permite até $2m$ conexões por nó⁴³. Essa duplicação de limites na base é implementada para garantir a conectividade global do grafo e evitar que agrupamentos isolados de dados fiquem inacessíveis à travessia, mantendo o grafo resiliente a remoções e minimizando a existência de mínimos locais⁴³. O valor padrão no pgvector é m (com variação típica entre m e $2m$)⁵.

O ajuste desses parâmetros exige a compreensão de um compromisso triplo entre tempo de compilação, consumo de memória RAM e taxa de revocação (*recall*) das consultas⁵.

Configuração de Parâmetros | Tempo de Build | Consumo de RAM | Taxa de Recall | QPS (Consultas/seg)

Sintonia Fina de hnsw.efl_search em Tempo de Execução

Diferente dos parâmetros de criação que são imutáveis após a compilação do índice, o parâmetro hnsw.efl_search (padrão 100) atua em tempo de execução para controlar a largura da busca gananciosa na Camada 06. Ele define o tamanho máximo da fila de prioridades que mantém os candidatos a vizinho mais próximo durante o rastreamento final⁸.

* **Priorização de Vazão (QPS):** Configurar hnsw.efl_search para valores menores (ex: 10 ou 5) reduz o número total de cálculos de distância por query⁵. Essa configuração é indicada para sistemas de alto tráfego concorrente, como preenchimento automático de busca, onde a latência de milissegundos únicos é mais importante do que encontrar todas as correspondências semânticas perfeitas³³.

Administração Física e Ajuste de Performance (Tuning)

Dimensionamento Físico de Memória RAM para Índices HNSW

Diferente de índices tradicionais como B-Tree, o HNSW exige que a sua estrutura de nós e listas de adjacência resida integralmente na memória RAM do servidor para evitar a degradação de desempenho²⁸. Devido ao padrão aleatório de travessia do grafo, qualquer necessidade de paginação física para ler blocos de disco (*page faults*) eleva a latência da consulta por ordens de grandeza²⁸.

*  é o volume total de vetores indexados48.

A tabela a seguir apresenta os requisitos de memória estimados sob diferentes dimensionalidades e volumes de dados para orientar o provisionamento de hardware, adicionando uma margem de segurança de 20% recomendada para acomodar o overhead de páginas físicas do PostgreSQL e o cache operacional²³.

Volume de Vetores (N) | Dimensão (d) | Conexões (m) | Tamanho Bruto do Índice | RAM Mínima Recomendada

Parâmetros Recomendados para o postgresql.conf

A otimização de consultas vetoriais complexas exige o ajuste de parâmetros globais do PostgreSQL para assegurar a eficiência de cache do grafo e paralelizar tarefas de infraestrutura²⁰.

Ini, TOML

O parâmetro `maintenance\work\mem` desempenha papel crítico na velocidade de criação do índice6. Durante a execução do comando `CREATE INDEX ... USING hnsw`, o PostgreSQL aloca as estruturas dinâmicas de busca gananciosa diretamente neste segmento de memória41. Se o volume de dados exceder o limite configurado, o PostgreSQL emitirá a seguinte mensagem de alerta no log6:

NOTICE: hnsw graph no longer fits into maintenance\work\mem after X tuples

Este aviso indica que o grafo ultrapassou o espaço em memória reservado e que o motor do banco de dados iniciou a gravação de arquivos temporários em disco, o que causa degradação no tempo de construção do índice6. Para cargas acima de 1 milhão de vetores de 1536 dimensões, recomenda-se configurar `maintenance\work\mem` para no mínimo 4 GB durante a execução do `reindex6`.

Estratégias de Manutenção Física de Índices Vetoriais

À medida que novas linhas são inseridas, atualizadas ou excluídas do banco de dados, os índices vetoriais sofrem degradação estrutural silenciosa²⁸.

Protocolo de Manutenção Recomendado para DBAs

*** Monitoramento de Desempenho e Coleta de Métricas:** Estabelecer uma rotina de monitoramento estatístico periódico, medindo a taxa de acerto de cache de buffer de bloco por meio das tabelas de sistema do PostgreSQL (`pg\_statio\_user\_indexes`)⁵². Se a taxa de acertos cair abaixo de  ⁵⁶, significa que o índice HNSW ultrapassou os limites saudáveis de RAM e está executando paginação física em disco⁵².

SQL

O PostgreSQL aloca processos paralelos em background para construir um grafo HNSW espelhado e completamente novo com base nos dados consolidados mais recentes da tabela, sem suspender as queries simultâneas e escritas da aplicação de produção²⁸. Uma vez finalizado o processamento físico e validadas as conexões, as referências de catálogo são atualizadas atômicamente e o grafo antigo fragmentado é liberado do pool de páginas físicas em disco de forma transparente²⁸.

Proposta Prática e Arquitetura de Demonstração (RAG)

Para garantir o isolamento físico de dados sensíveis entre clientes de uma plataforma multi-inquilino (*multi-tenant*) e evitar o vazamento de informações contextuais em pipelines de RAG, projeta-se abaixo a arquitetura de banco de dados e backend¹².

\+-----+

Script de Implementação do Banco de Dados (SQL)

SQL

-- Tabela principal para armazenamento dos Documentos Mestre (Metadados Globais)

-- Tabela especializada para retenção de fragmentos de texto (Chunks) e embeddings associados

\-- Geração inline e dinâmica de coluna tsvector em português para busca full-text lexical

-- Restrição de unicidade para evitar inserções em duplicidade do mesmo fragmento

-- Indexação clássica relacional B-Tree para otimizar filtros em nível de chave de inquilino

-- Criação do índice vetorial HNSW com otimização estrutural de barreira de classe

**-- Indexação lexical GIN na coluna gerada para otimizar pesquisas baseadas em palavras-chave
[cite: 60]**

Integração do Backend de Aplicação (Python \+ Psycopg 3\)

O script abaixo demonstra de forma clara como estruturar o processamento no backend da aplicação utilizando a biblioteca `psycopg` para interagir diretamente com o PostgreSQL⁶¹. O backend é encarregado de extrair textos, obter os **embeddings** de alta dimensionalidade via chamadas locais de NLP e acionar as queries de busca híbrida com busca iterativa ativa para alimentar a inferência do LLM¹¹.

Python

```
class RetrievalPipeline:
```

```
def _get_embedding(self, text: str) -> np.ndarray:
```

```
def execute_hybrid_search(self, tenant_id: str, query_text: str, limit: int = 5) -> list:
```

```
results := \[
```

```
with conn.cursor() as cur:
```

\# Executa a query híbrida unificando os filtros de metadados relacionais e ordenação vetorial

```
cur.execute(hybrid_query, (query_vector, tenant_id, query_vector, limit))
```

```
for row in rows:
```

```
def generate_rag_context(self, tenant_id: str, query_text: str) -> str:
```

```
\# Consolida os dados recuperados em um bloco de texto unificado
```

```
full\_context \= "\n".join(context\_blocks)
```

```
\# Constrói o prompt final blindado para alimentação semântica do LLM downstream \[cite: 11, 66\]
```


Compilação de Benchmarks e Casos de Estudo

Dados Empíricos de Desempenho

Os benchmarks compilados nesta seção refletem o comportamento observado sob cargas sintéticas de *embeddings* de alta dimensionalidade (1024 dimensões padrão float32, compatíveis com a infraestrutura das APIs clássicas de IA) executados sob condições otimizadas de hardware de nuvem em instâncias com SSD NVMe local e barramento dedicado de memória²³.

1. Busca Sequencial (KNN Exato): Sem uso de índices vetoriais35.

Tamanho da Tabela (Vetores) | Algoritmo de Indexação | Parâmetros de Execução | Latência p50 (ms) | Latência p99 (ms) | Taxa de Recall (%) | Throughput (QPS)

Análise de Desempenho e Comportamento de Escala

A análise empírica revela que a busca sequencial exata exibe um comportamento estritamente linear no tempo de execução, escalando de  ms na escala de 10k para proibitivos  ms quando a base alcança 1 milhão de vetores³⁵. Esse tempo inviabiliza qualquer uso em aplicações em tempo real, onde as restrições de tempo de resposta exigem latências de consulta inferiores a 50 ms⁵¹. Consequentemente, a busca sem índices torna-se impraticável para volumes de dados acima de 50.000 registros, justificando o uso de algoritmos de aproximação¹³.

A Transição Arquitetural para Escalas Avançadas (Mais de 10M de Vetores)

Conclui-se que o HNSW de base puramente em RAM apresenta excelente desempenho para bases de dados contendo até 5 a 10 milhões de vetores, desde que o servidor possua capacidade financeira para acomodar o tamanho do índice totalmente residente nos buffers do SGBD²³.

Referências Bibliográficas

* KANE, Andrew. ***pgvector***: Open-source vector similarity search for Postgres. GitHub Repository, 2026\. Disponível em: [\[https://github.com/pgvector/pgvector\]](https://github.com/pgvector/pgvector)(<https://github.com/pgvector/pgvector>).6

Referências citadas

1. PGVector: Transforming PostgreSQL into a Powerful Vector Database for AI Applications | by Amit Dhiman | Medium, https://medium.com/@amitt dhiman91/pgvector-transforming-postgresql-into-a-powerful-vector-database-for-ai-applications-d2716689b50d

[image1]:

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[image2]:

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